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CAAENet: Cosine-Driven Adversarial Auto-Encoder Network for Hyperspectral Image Classification

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Abstract—Various deep learning networks represented by auto-encoders (AEs) have obtained numerous achievements in hyperspectral image (HSI) classification. However, most of them have no consideration of feature distribution and separability, so that make it limited for the obtained features to be distinguished by the prediction layer. To remedy such deficiency, in this study, we propose a cosine-driven adversarial auto-encoder network (CAAENet) for HSI classification, which integrates the adversarial style and cosine representation into the AE framework. More concretely, we firstly treat the AE network as the basic framework. Then, the adversarial loss is constructed into the basic AE framework, which aims to increase the distribution consistency between the input HSIs and their features by discriminator during adversarial learning. Thirdly, we build the cosine loss into the basic network to improve the feature separability, so that the intra-class features are aggregated and the inter-class distance is enlarged. Fourthly, an alternating optimization strategy is designed to obtain the optimal solutions of the overall loss of CAAENet. Finally, the label prediction of the input HSIs is achieved by imposing the one-hot encoding on a series of the calculation of the obtained solutions. Different from most of existing relevant approaches, the proposed CAAENet method takes full consideration of feature distribution and separability to capture the highly separable and discriminative features. Extensive experiments have been conducted to demonstrate that the proposed CAAENet method achieves more competitive performance compared with several representative state-of-the-art approaches.

Index Terms—hyperspectral image classification, distribution, separability, auto-encoders, cosine loss, adversarial loss

I. INTRODUCTION

Hyperspectral image (HSI) classification has emerged as a pivotal task in remote sensing. Unlike traditional multispectral or natural images, hyperspectral data captures hundreds of continuous spectral bands [1], offering fine-grained spectral information about materials on the Earth's surface. This rich spectral resolution enables the discrimination of subtle differences among various land cover types, such as different crop species or urban materials. Accurate HSI classification is thus vital for applications like environmental monitoring, precision agriculture, mineral exploration, and disaster management, facilitating informed decision-making in diverse fields [2]–[6].

In the early years, many works have mainly focused on the shallow features of the input HSIs. Typical morphology [7]–[9] and representation learning [10]–[12] are here for examples. The morphology manner analyzes object shapes included in HSIs by using structural elements to perform transformations like dilation and erosion for feature extraction. Benediktsson et al. [7] have presented the famous multi-scale morphology based method combines the geodesic opening

and closing operations so that a differential morphological profile is constructed to describe the structural information. Based on such work product, many scholars have also designed other extended morphology versions such as extended morphological attribute profiles (EMAP) [8] and directional morphological profiles (DMP) [9]. More commonly, the representation learning is an interesting and effective manner, which is often achieved by introducing sparse representation (SR) and low-rank representation (LRR). According to the SR theory, it is assumed that an unknown sample is approximately represented by the combination of a few training samples from the entire dictionary whereas the corresponding sparse vector encodes the class information implicitly [13], such as L1-norm-regularization [14], Laplacian regularized Lasso [15], group sparse coding [16], and joint sparsity [17]. While the LRR considers that a particular form of the given HSI under the lowest-rank constraint is found to obtain the global structure of data [18]. For example, the low-rank discriminative least squares regression model (LRDLRSR) [19] and the diagonalized low-rank Learning model (DLRL) [20] class-wisely impose the low-rank constraint to mine structural characteristics.

In the recent decade, deep learning [21]–[27] has achieved significant progress in the HSI classification, which pays close attention to deep-level features of HSIs. Compared with traditional shallow features, deep-level features are potential to capture complex patterns and representations within the input HSI data. As can be widely known, the convolutional neural network (CNN) [28]–[30] is currently the most commonly used deep learning architecture and has achieved success in HSI classification. For example, Cao *et al.* [28] propose an integrated network framework (CNN-AL), which introduces the active learning into the CNN framework. In [29], the squeeze multibias network (SMBN) is proposed, which uses an efficient squeeze convolution module to efficiently enhance the running speed of the network by reducing the number of parameters. In [30], an efficient nonlocal module based deep fully CNN (ENL-FCN) is designed, which incorporates the long-range contextual information for HSI classification.

During the past about five years, the Transformer model [31]–[34] has become very impressive in the task of HSI classification. As a novel type of networks, the Transformer leverages self-attention mechanisms to capture global dependencies between elements in HSI data, which enables parallel processing while uses positional encoding to retain sequence order information for efficient long-range context modeling. Graham *et al.* [32] have designed a named LeViT method, where the attention bias is introduced to integrate positional information in vision transformers for the purpose

of classification. In [33], a hyperspectral image transformer (HiT) classification network is proposed by embedding convolution operations into the transformer structure to capture the subtle spectral discrepancies and convey the local spatial context information. In [34], the self-supervised masked image reconstruction is introduced to advance transformer models for the classification of hyperspectral remote sensing imagery.

In addition, the auto-encoders (AEs) [35]–[38] are another commonly focused network structure in the HSI classification task. This type aims to learn efficient hyperspectral data representations by training a neural network to compress input into a lower-dimensional latent space (encoding) and then reconstruct it (decoding), minimizing reconstruction error to capture essential features. Chen et al. [35] have for the first time presented the AE based method for HSI classification, which introduces the stacked auto-encoder (SAE) to obtain discriminative deep-level features used for label prediction. In [36], a framework called compact and discriminative SAE (CDSAE) has been designed, which can learn discriminative low dimensional feature mappings and train an effective classifier progressively. In [38], a tensored deep autoencoder (TDAE) is proposed for classification of HSIs, where the tensor decomposition constraint item is built into a deep auto-encoders network with a fully connection layer, so as to achieve the integration of multi-layer output features and capture the structure information among outputs. Besides these common CNNs, Transformers, and AEs, there exist other typical methods for HSI classification. For example, Xu et al. [39] developed a robust self-ensembling network (RSEN) to address the classification task. Chen et al. [40] devised a deep belief network (DBN) to extract and leverage spectral-spatial features for HSI classification.

It is noteworthy that most of networks discussed-above for HSI classification have no consideration of feature distribution and separability. On the one hand, when training a given network, features within HSIs are learned, and then used to train a classifier like a soft-max layer for result prediction. The implicit feature space does not consider the distribution, which produces the distribution shift between the input HSIs and their features. Therefore, how to increase the distribution consistency between the input HSIs and their features is the *first motivation* of this work. On the other hand, from the perspective of the geometric feature display, most existing methods have no explicit concerns on the separability in feature space. Actually, the high separability should be achieved by aggregating intra-class features and increasing inter-class distances. Based on such argument, how to achieve the high separability of features is the *second motivation* of our work.

For the sake of embracing both motivations pointed-above, in this work, we propose a novel cosine-driven adversarial auto-encoder network (CAAENet) for HSI classification. More specifically, we firstly consider using the AE network as the basic framework. Then, in order to increase the distribution consistency between the input HSIs and their features, the adversarial loss is constructed into the AE network. The distribution difference is continuously reduced by the discriminator during the adversarial training. Thirdly, considering the geometric feature display, the cosine loss is also built into the basic

network to improve the separability in the feature space. By such way, the intra-class features are aggregated and the inter-class distance is enlarged, which facilitates the classification calculation. Fourthly, to solve the overall loss of CAAENet, an alternating optimization strategy is formulated for obtaining optimal solutions. Finally, the label prediction of the input HSIs is achieved by imposing the one-hot encoding on a series of the calculation of the obtained solutions. Compared with most of existing approaches, our proposed CAAENet method takes full consideration of both feature distribution and separability, so as to capture the highly separable and discriminative features for the high-performance HSI classification. Extensive experimental results have been given to demonstrate that the proposed CAAENet method achieves the better performance than several representative advanced approaches in the HSI classification task.

The main novel contributions of this study are summarized into the four-fold pieces. *First*, we propose the CAAENet method for HSI classification, which integrates the adversarial style and cosine representation into the AE framework. *Second*, we construct the adversarial loss introduced into the basic AE network, which aims to guarantee the distribution consistency between the input HSIs and their features. *Third*, we build the cosine loss into the basic network, which is devoted to improving the separability in the feature space for convenient discriminant-analysis. *Fourth*, we design an alternating optimization strategy to acquire the optimal solutions of the overall loss of the CAAENet method.

The rest of this article is organized below. The proposed CAAENet method is introduced in Section II. The experiment is given in Section III. The conclusion is finally presented in Section IV.

II. METHODOLOGY

In this section, we will introduce the details of the proposed CAAENet method for HSI classification. We here give some notes: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ is the input HSI dataset, where $x_i \in \mathbb{R}^{d_L}$ is the pixel and $y_i \in \{1, 2, \dots, C\}$ denotes the corresponding label. $W = \{w_1, w_2, \dots, w_C\} \in \mathbb{R}^{d_F \times C}$ is the weight matrix for classification, where w_i denotes the weight vector for the i -th class.

A. Motivation Analysis and Model Formulation

1) *Motivation Analysis*: As mentioned-previously, AEs provide an effective solution to the problem of the HSI classification through their powerful ability for unsupervised feature learning. The AE forms are neural networks trained to reconstruct their input, which can capture the most essential and discriminative features within HSIs while filtering out noise and redundancy. These learned features are then used to train a classifier like a soft-max layer for result prediction. It is noted that the implicit feature space has no consideration of the distribution, which produces the distribution shift between the input HSI and its features. Fortunately, some evidences [41]–[43] have demonstrated that the adversarial learning can effectively deal with the distribution shift problem. The adversarial style does not directly modify the data or model to adapt

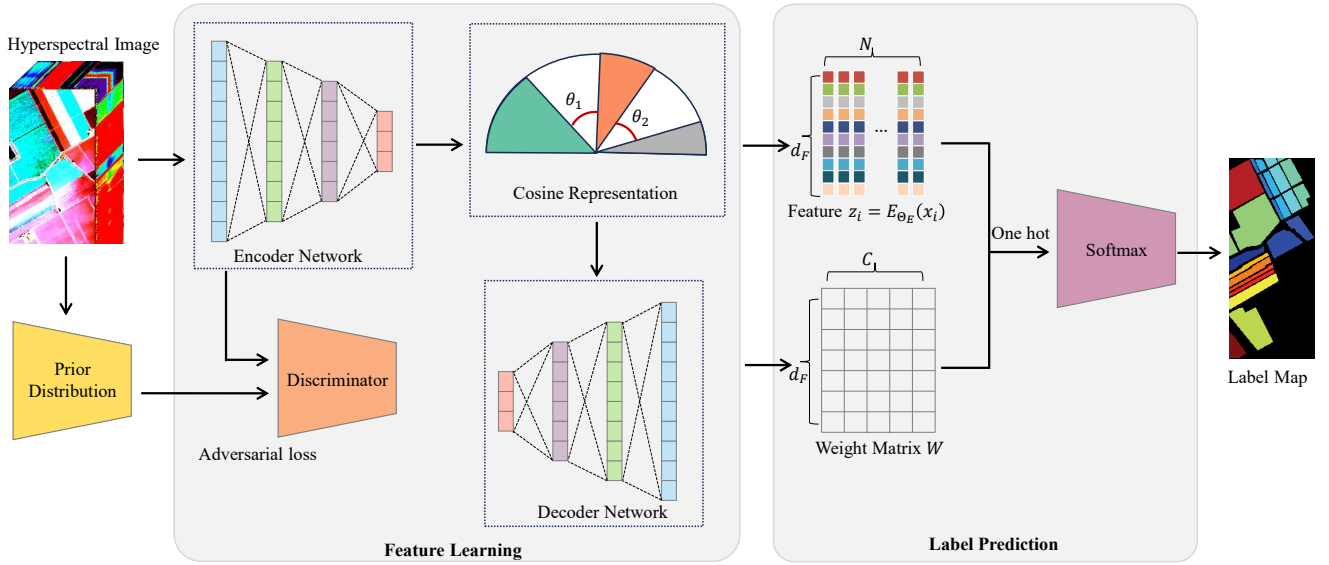


Fig. 1. Overview of the proposed CAAENet method for HSI classification, which integrates the adversarial style and cosine representation into the auto-encoder framework.

to a specific shift, but force the model to learn similar feature distributions between domains by an inherent and game training mechanism.

In addition, from the perspective of the geometric feature display, existing methods often do not focus on the separability in feature space. For example, end-to-end deep learning methods [36], [40], which are the most used network frameworks, achieve the HSI classification by jointly optimizing the feature extraction layers and prediction layer. Actually, the good separability should be achieved by aggregating intra-class features and increasing inter-class distances. Cosine space can provide an effective and interpretable metric form that only focuses on vector direction (class directivity) without considering strength. Therefore, cosine representation has great potential to achieve the high separability and discriminability for improving classification performance.

2) *Model Formulation:* According to the analyzed-above, taking the consideration of both feature distribution and geometric feature display, this work proposes a cosine-driven adversarial auto-encoder network (CAAENet) shown in Fig. 1 for HSI classification, which integrates the adversarial style and cosine representation into the AE framework. The overall loss function of CAAENet should include at least three parts, *i.e.*, reconstruction loss for AE, adversarial Loss, and cosine loss. Based on such claim, the mathematical model of CAAENet can be constructed as:

$$L_{overall} = L_{recon} + \lambda_1 L_{adv} + \lambda_2 L_{cos}, \quad (1)$$

where $L_{overall}$ represents the overall loss of our proposed CAAENet; L_{recon} is the reconstruction loss for AE; L_{adv} is the adversarial Loss used to guarantee the distribution consistency between the input HSIs and their features; L_{cos} denotes the cosine loss used to achieve the high separability of features; λ_1 and λ_2 denote the coupling parameters. According

to the AE theory [44], the reconstruction loss can be written into:

$$L_{recon}(\Theta_E, \Theta_D) = \frac{1}{N} \sum_{i=1}^N \|x_i - D_{\Theta_D}(E_{\Theta_E}(x_i))\|_F^2, \quad (2)$$

where E_{Θ_E} and D_{Θ_D} are respectively the encoder and decoder of AE, whose network parameter are denoted by Θ_E and Θ_D . As can be found in (2), the encoder E_{Θ_E} is used to map the input x_i into the feature $z_i \in \mathbb{R}^{d_F}$ in the corresponding implicit space. The decoder D_{Θ_D} is employed to reconstruct feature z_i into $\hat{x}_i \in \mathbb{R}^{d_L}$. In summary, this reconstruction loss ensures that the encoded features preserve the essential information of the original HSIs, enabling the decoder to recover the input with minimal distortion. Consequently, the latent representations become reliable for downstream tasks. However, without additional constraints, the feature space may lack distribution consistency and class separability, which motivates the introduction of adversarial and cosine losses.

Referring to the adversarial learning theory [45], the adversarial learning is achieved by alternating optimization of both generator and discriminator. Considering the AE framework, the function of generator is realized via the encoder E_{Θ_E} shown in (2). Thus, the adversarial loss of the encoder is formulated into:

$$L_{adv}^E(\Theta_E, \Theta_P) = \frac{1}{N} \sum_{i=1}^N [-\log P_{\Theta_P}(E_{\Theta_E}(x_i))], \quad (3)$$

where L_{adv}^E denotes the adversarial loss of the encoder, aiming to update the encoder. P_{Θ_P} is the discriminator network, whose parameter set is defined as Θ_P . The input of P_{Θ_P} is map feature z_i , and the output is the probability from prior distribution. The adversarial loss of the discriminator can be formed as follows:

$$L_{adv}^P(\Theta_E, \Theta_P) = \frac{1}{N} \sum_{i=1}^N [\log P_{\Theta_P}(z_i^{prior}) + \log(1 - P_{\Theta_P}(E_{\Theta_E}(x_i)))] \quad (4)$$

where L_{adv}^P denotes the adversarial loss of the discriminator used to guarantee the distribution consistency. $z_i^{prior} \sim p_{prior}(z)$ denotes the prior distribution, which usually follows a standard normal distribution $\mathcal{N}(0, 1)$. By minimizing this discriminator loss, the network effectively distinguishes between real prior samples and encoded features, while the encoder simultaneously tries to fool the discriminator. This adversarial game gradually aligns the feature distribution with the prior, reducing distribution shift. Consequently, the learned features become more consistent with the desired prior, improving generalization and robustness in classification tasks.

Furthermore, we have claimed that the cosine representation is feasible to achieve the discriminability of features. The modified normalized SoftMaxloss [46] is available to complete the cosine representation, which can be written as:

$$L_{cos}(\Theta_E, W) = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i})}}{\sum_{j=1}^N e^{s \cos(\theta_{y_j})}}, \quad (5)$$

where s is a scale factor for more stable computation. $\theta_i = w_i^T z_j$ is the cosine similarity, which reflects the angle between both w_i and z_j , so as to promote the exported features from the same class to be around an axis. Here considering the case of 2-D space for example, the feature vector z_i is mapped into location near one of two axes. If z_i is located near axis-1, there exists $\cos(\theta_1) > \cos(\theta_2)$. On the contrary, $\cos(\theta_1) < \cos(\theta_2)$. In order to achieve a more discriminative classification boundary, a margin constant m is introduced, i.e.,

$$L_{cos_f}(\Theta_E, W) = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} - m)}}{\sum_{y_j \neq y_i} e^{s \cos(\theta_{y_j})} + e^{s \cos(\theta_{y_i} - m)}}. \quad (6)$$

With Eq. (6), the margin constant m is introduced so as to satisfy $\cos(\theta_1) - m > \cos(\theta_2)$ or $\cos(\theta_1) < \cos(\theta_2) - m$ in the case of 2-D space, which is indeed to enhance the discriminability. As can be found, the margin m is fixed, which may obtain a sufficient representation due to the highly biased distribution of different categories. To remedy such deficiency, we introduce the adaptive margin m_i into the model as following:

$$L_{cos_a}(\Theta_E, W) = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} - m_i)}}{\sum_{j=1}^N e^{s \cos(\theta_{y_i} - m_j)}}, \quad (7)$$

where $m_i = 1 - \frac{n_i + \epsilon}{\sum_{j=1}^N n_j + \epsilon}$ is a adaptive margin for x_i . ϵ is a very small constant. The adaptive margin m_i adjusts the angular penalty per sample based on its class characteristics, which effectively mitigates the class imbalance issue and enhances the intra-class compactness for categories with large spectral diversity. As a result, the learned features become more discriminative and robust, leading to improved classification accuracy in complex hyperspectral scenes.

Considering the integration of Eqs. (2), (3), and (7), the overall loss of CAAENet in (1) can be finally updated as

following:

$$L_{overall}(\Theta_E, \Theta_D, \Theta_P, W) = L_{recon}(\Theta_E, \Theta_D) + \lambda_1 L_{adv}^E(\Theta_E, \Theta_P) + \lambda_2 L_{cos_a}(\Theta_E, W). \quad (8)$$

Once the overall loss function in Eq. (8) is minimized, the encoder parameters Θ_E and the classification weight matrix W become well adapted to both the adversarial distribution alignment and the cosine margin constraint. This ensures that the learned features are not only consistent with the input distribution but also highly separable across different classes, thereby boosting the final classification accuracy. Next, we provide the altering optimization procedure.

B. Alternating Optimization Strategy

The formula (8) includes multiple variables, i.e., Θ_E , Θ_D , Θ_P , and W , which is non-convex. Therefore, we design an alternating update optimization strategy to solve the optimal solution of formula (8). The detailed procedure is seen below.

1) *Update Optimization of Θ_P* : The E_{Θ_E} and P_{Θ_P} are the encoder and discriminator. According to the adversarial learning theory [45], the encoder and discriminator are alternately optimized to determine Θ_E and Θ_P . In order to obtain update Θ_P , we fix the encoder E_{Θ_E} in (4) as following:

$$\min_{\Theta_P} L_{adv}^P(\Theta_E, \Theta_P). \quad (9)$$

We here introduce the gradient descent scheme [47], [48] to obtain the updated Θ_P , that is:

$$\Theta_P = \Theta_P - \eta_P \frac{\partial L_{adv}^P(\Theta_E, \Theta_P)}{\partial \Theta_P}, \quad (10)$$

where η_P is the learning rate. $\frac{\partial L_{adv}^P(\Theta_E, \Theta_P)}{\partial \Theta_P}$ represents the partial derivative of $L_{adv}^P(\Theta_E, \Theta_P)$ with respect to Θ_P , which can be acquired by:

$$\begin{aligned} & \frac{\partial L_{adv}^P(\Theta_E, \Theta_P)}{\partial \Theta_P} \\ &= \frac{1}{N} \sum_{i=1}^N \left[\frac{\partial \log P_{\Theta_P}(z_i^{prior})}{\partial \Theta_P} + \frac{\partial \log(1 - P_{\Theta_P}(E_{\Theta_E}(x_i)))}{\partial \Theta_P} \right]. \end{aligned} \quad (11)$$

2) *Update Optimization of W* : Fixing Θ_E , Θ_D , and Θ_P , the overall loss in formula (8) is turned into the loss form only regarding to W , i.e.,

$$\min_W L_{cos_a}(\Theta_E, W). \quad (12)$$

The gradient descent scheme is also used to update the weight matrix W :

$$W = W - \eta_W \frac{\partial L_{cos_a}(\Theta_E, W)}{\partial W}, \quad (13)$$

where η_W is the learning rate. $\frac{\partial L_{cos_a}(\Theta_E, W)}{\partial W}$ is the partial derivative of $L_{cos_a}(\Theta_E, W)$ with respect to W . Considering $v_i = \log \frac{e^{s \cos(\theta_i - m_i)}}{\sum_{j=1}^N e^{s \cos(\theta_j - m_j)}}$, $\theta_i = w_i^T z_j$, and $W =$

$\{w_1, w_2, \dots, w_C\}$, the partial derivative can be written as:

$$\frac{\partial L_{\cos a}(\Theta_E, W)}{\partial w_i} = s \times (v_i - 1\{i = y_j\}) \times \frac{\partial \cos(\theta_i)}{\partial w_i}, \quad (14)$$

where $1\{i = y_j\}$ is the indicator function, and $\frac{\partial \cos(\theta_i)}{\partial w_i}$ is the partial derivative of $\cos(\theta_i)$ for w_i . Considering the normalization, such derivative is described into:

$$\begin{aligned} \frac{\partial \cos(\theta_i)}{\partial w_i} &= \frac{\partial}{\partial w_i} \left(\frac{w_i^T z_j}{\|w_i\|_2 \|z_j\|_2} \right) \\ &= \frac{1}{\|w_i\|_2 \|z_j\|_2} \left(I - \frac{w_i w_i^T}{\|w_i\|_2^2} \right) z_j. \end{aligned} \quad (15)$$

3) *Update Optimization of Θ_E* : Keeping Θ_D , Θ_P , and W fixed, the overall loss in (8) is relaxed into the optimization formula with respect to Θ_E below:

$$\min_{\Theta_E} L_{\text{overall}}(\Theta_E, \Theta_D, \Theta_P, W). \quad (16)$$

We calculate the partial derivative of L_{overall} for Θ_E as:

$$\begin{aligned} \frac{\partial L_{\text{overall}}}{\partial \Theta_E} &= \frac{\partial L_{\text{recon}}(\Theta_E, \Theta_D)}{\partial \Theta_E} + \lambda_1 \frac{\partial L_{\text{adv}}^E(\Theta_E, \Theta_P)}{\partial \Theta_E} \\ &\quad + \lambda_2 \frac{\partial L_{\cos a}(\Theta_E, W)}{\partial \Theta_E}, \end{aligned} \quad (17)$$

where $\frac{\partial L_{\text{recon}}(\Theta_E, \Theta_D)}{\partial \Theta_E}$, $\frac{\partial L_{\text{adv}}^E(\Theta_E, \Theta_P)}{\partial \Theta_E}$, and $\frac{\partial L_{\cos a}(\Theta_E, W)}{\partial \Theta_E}$ are respectively the partial derivative of $L_{\text{recon}}(\Theta_E, \Theta_D)$, $L_{\text{adv}}^E(\Theta_E, \Theta_P)$, and $L_{\cos a}(\Theta_E, W)$ for Θ_E . Due to $E_{\Theta_E}(x_i) = z_i$, these three partial derivative can be specifically described into:

$$\begin{aligned} \frac{\partial L_{\text{recon}}(\Theta_E, \Theta_D)}{\partial \Theta_E} &= \frac{2}{N} \sum_{i=1}^N (x_i - D_{\Theta_D}(E_{\Theta_E}(x_i))) \\ &\quad \times \frac{\partial D_{\Theta_D}(E_{\Theta_E}(x_i))}{\partial E_{\Theta_E}(x_i)} \cdot \frac{\partial E_{\Theta_E}(x_i)}{\partial \Theta_E}, \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{\partial L_{\text{adv}}^E(\Theta_E, \Theta_P)}{\partial \Theta_E} &= -\frac{1}{N} \sum_{i=1}^N \frac{1}{P_{\Theta_P}(E_{\Theta_E}(x_i))} \\ &\quad \times \frac{\partial P_{\Theta_P}(E_{\Theta_E}(x_i))}{\partial E_{\Theta_E}(x_i)} \cdot \frac{\partial E_{\Theta_E}(x_i)}{\partial \Theta_E}, \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{\partial L_{\cos a}(\Theta_E, W)}{\partial \Theta_E} &= -\frac{1}{N} \sum_{i=1}^N s \cdot \left(\sum_{j=1}^C (1\{j = y_i\} - v_i) \right. \\ &\quad \times \left. \frac{\partial \cos(\theta_j)}{\partial z_i} \right) \times \frac{\partial E_{\Theta_E}(x_i)}{\partial \Theta_E}, \end{aligned} \quad (20)$$

where $\frac{\partial \cos(\theta_j)}{\partial z_i} = \frac{1}{\|z_i\|_2} \left(I - \frac{z_i z_i^T}{\|z_i\|_2^2} \right)$. Based on the above, the gradient descent update is achieved by:

$$\Theta_E = \Theta_E - \eta_E \frac{\partial L_{\text{overall}}}{\partial \Theta_E}, \quad (21)$$

where η_E denotes the learning rate.

4) *Update Optimization of Θ_D* : Fixing Θ_E , Θ_P , and W , Θ_D can be updated by solving the following form:

$$\min_{\Theta_D} L_{\text{recon}}(\Theta_E, \Theta_D). \quad (22)$$

Similarly, we compute its partial derivative with respect to Θ_D below:

$$\begin{aligned} \frac{\partial L_{\text{recon}}(\Theta_E, \Theta_D)}{\partial \Theta_D} &= \frac{2}{N} \sum_{i=1}^N (x_i - D_{\Theta_D}(E_{\Theta_E}(x_i))) \\ &\quad \times \frac{\partial D_{\Theta_D}(E_{\Theta_E}(x_i))}{\partial \Theta_D}. \end{aligned} \quad (23)$$

With the gradient descent strategy, the Θ_E can be updated by:

$$\Theta_D = \Theta_D - \eta_D \frac{\partial L_{\text{recon}}(\Theta_E, \Theta_D)}{\partial \Theta_D}, \quad (24)$$

where η_D denotes the learning rate.

The solutions of (8) are obtained by the update optimization in Eqs. (10), (13), (21), and (24), once the update iteration stops. The stop conditions are as follows:

$$\begin{cases} \|\Theta_P^{t+1} - \Theta_P^t\|_2 \rightarrow 0, \\ \|W^{t+1} - W^t\|_2 \rightarrow 0, \\ \|\Theta_E^{t+1} - \Theta_E^t\|_2 \rightarrow 0, \\ \|\Theta_D^{t+1} - \Theta_D^t\|_2 \rightarrow 0, \end{cases} \quad (25)$$

where Θ_P^t , W^t , Θ_E^t , and Θ_D^t are respectively the outputs at the t -th iteration update.

C. Label Prediction of The input HSIs

In our proposed CAAENet, the one-hot encoding [49] is introduced to describe the prediction label information. The prediction output with the probability distribution is yielded by the soft-max function. Thus, for an unknown sample $\tilde{x}$, its prediction probability distribution $\tilde{y}$ can be determined via:

$$\begin{aligned} \tilde{y} &= \text{softmax}(\mathbf{o}) \\ &= \left[\frac{e^{o_1}}{\sum_{j=1}^C e^{o_j}}, \frac{e^{o_2}}{\sum_{j=1}^C e^{o_j}}, \dots, \frac{e^{o_C}}{\sum_{j=1}^C e^{o_j}} \right]^T, \end{aligned} \quad (26)$$

where $\mathbf{o} = [o_1, o_2, \dots, o_C]^T$ is the output logits of the last layer of the model, where $o_j = s \cdot \cos(\theta_j) = w_j^T E_{\Theta_E}(\tilde{x})$. Thus, the final classification decision is expressed into:

$$\tilde{y} = \max_j \tilde{y}_j, \quad (27)$$

where $\tilde{y}_j = \frac{e^{o_j}}{\sum_{j=1}^C e^{o_j}}$.

We here summarily remark four-fold points: *First*, for the CAAENet method, the adversarial style and cosine representation are integrated into the given AE framework. *Second*, the adversarial loss is constructed to guarantee the distribution consistency between the input HSIs and their features. *Third*, the cosine loss is built to improve the separability of features for convenient discriminant-analysis. *Fourth*, an alternating optimization strategy is designed in order to acquire the optimal solutions of the overall loss of the CAAENet method.

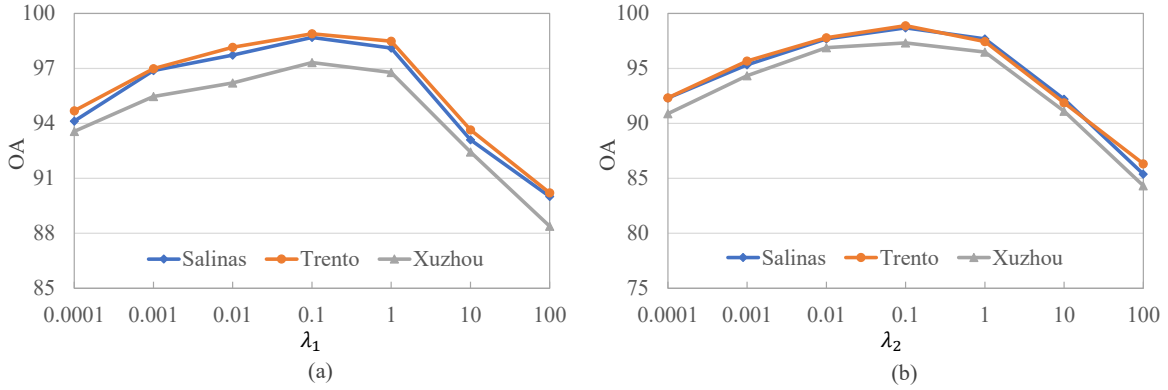


Fig. 2. The classification performance of our proposed CAAENet method influenced by two parameters (i.e., λ_1 and λ_2). (a) for λ_1 ; (b) for λ_2 .

III. EXPERIMENT

In the following, we will present the experiment study to empirically explore the performance of the proposed CAAENet method, including experimental preparation, parameter analysis, algorithm comparison, and ablation study.

A. Experimental Preparation

1) *Data Sets*: For the sake of the performance evaluation, three datasets, including Salinas, Trento, and Xuzhou, used in the following experimental study. The Salinas data was also collected by the 224-band Airborne/Visible Infrared Imaging Spectrometer (AVIRIS) system over Salinas Valley at California, where it also covers 16 classes. We have removed 20 water absorption bands and down-sampled spatial resolution with a factor of 2 to obtain the data form with the size of $512 \times 217 \times 204$. The Trento dataset was captured over a rural region in the south of Trento, Italy. This HSI data were acquired through the Airborne Imaging Spectrometer for Applications (AISA) Eagle system for hyperspectral imaging Eagle sensor, which is equipped with 63 spectral bands covering the range from $0.42 \mu m$ to $0.99 \mu m$ and 600×600 spatial resolution at the $1m$ interval, where 6 classes are included. The Xuzhou data was acquired via an airborne HYSPEX hyperspectral camera over a suburban area of Xuzhou, China, which comprises 500×260 pixels with an exceptionally high spatial resolution of 0.73 meters per pixel. In the following experiment conduction, 436 spectral bands were used after removing noisy bands within the wavelength range of $415 nm$ to $2508 nm$. The data covers 9 classes, including crops, vegetation, man-made structures, and coal fields. For the sake of clear experimental observation, the ratio of pixels in each data is set to 10%, 1%, and 5% as training samples, and the rest for testing.

2) *Quantitative Metrics*: In what follows, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are used to quantitatively evaluate the classification results yielded by all methods. The OA measures the percentage of correctly classified pixels. The AA is the average percentage of correctly classified pixels for each class. The Kappa corrects the correct percentage of classified pixels.

B. Parameter Analysis

In order to well explore the performance of the CAAENet method for HSI classification, the parameter analysis is conducted. As is mentioned in Section II, the λ_1 and λ_2 are two coupling parameters, which aims to reflect the significance of the adversarial loss and cosine representation loss in Eq. (1). These two weight parameters have the significant impact on the performance of the CAAENet method for HSI classification. Fig. 2 is here adopted to present the classification performance variation of the CAAENet method adjusted by both parameters λ_1 and λ_2 , where the OA index is used for quantitative evaluation.

1) *Analysis of Parameter λ_1* : As shown in Fig. 2(a), the change curves of the classification performance of the CAAENet method affected by the λ_1 are visually shown, which are conducted on the Salinas, Trento, and Xuzhou datasets. From this figure, it can be directly seen that the curves of CAAENet achieves the largest values at $\lambda_1 = 0.1$. When $\lambda_1 < 0.1$ and $\lambda_1 > 0.1$, the curves decrease down sharply. The reason is that the too small values of λ_1 weaken the significance of adversarial loss, which has no idea to achieve the satisfactory feature distributions. On the contrary, over-large values of λ_1 breaks the function of the main AE network. Thus, the $\lambda_1 = 0.1$ is an optimal choice for the Salinas, Trento, and Xuzhou datasets.

2) *Analysis of Parameter λ_2* : Fig. 2(b) is adopted to present the performance curves of the CAAENet with the variation of the λ_2 values on the Salinas, Trento, and Xuzhou datasets. Similarly, as can be found from Fig. 2(b), the curves achieves the biggest OA values at $\lambda_2 = 0.1$. The curves decrease down sharply when $\lambda_2 < 0.1$ and $\lambda_2 > 0.1$. The reason is that, over-high weights of the cosine representation loss fail to better capture the highly separable and discriminative features, which is limited to enhance the classification performance. The too large values of λ_2 suffer from breaking function of the main network. Thence, the λ_2 value is optimally selected as 0.1 on the Salinas, Trento, and Xuzhou datasets.

C. Algorithm Comparison

1) *Competing Methods*: In this study, the proposed CAAENet classification method is compared with eight advanced approaches including RSEN [39], CNN-AL [28], ENL-

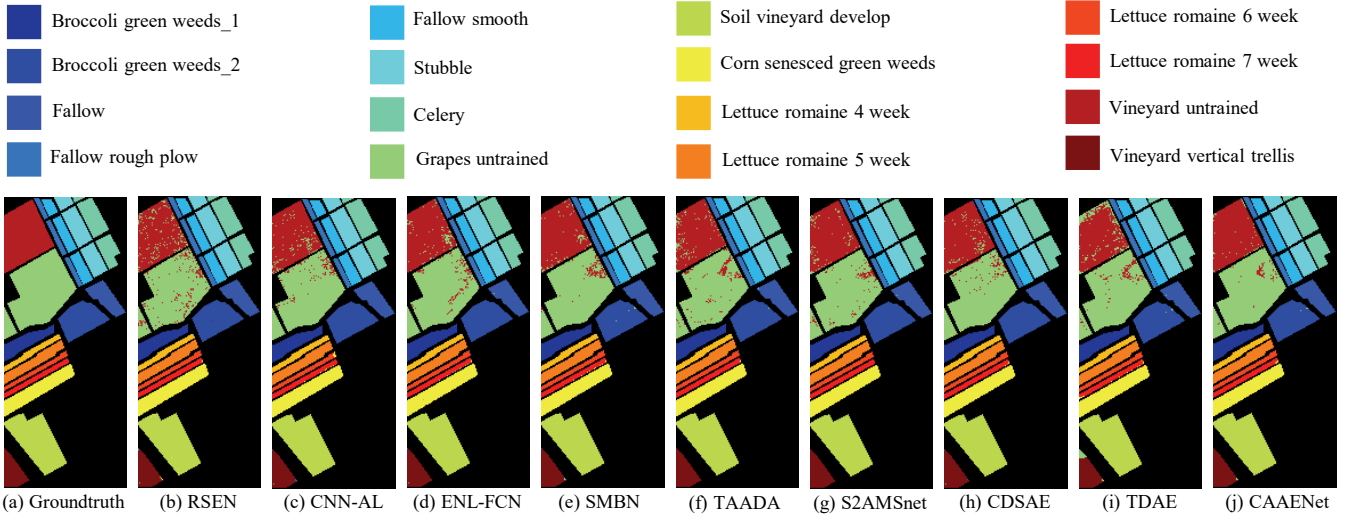


Fig. 3. Classification maps of different methods with 10% ratio of training samples on the Salinas data.

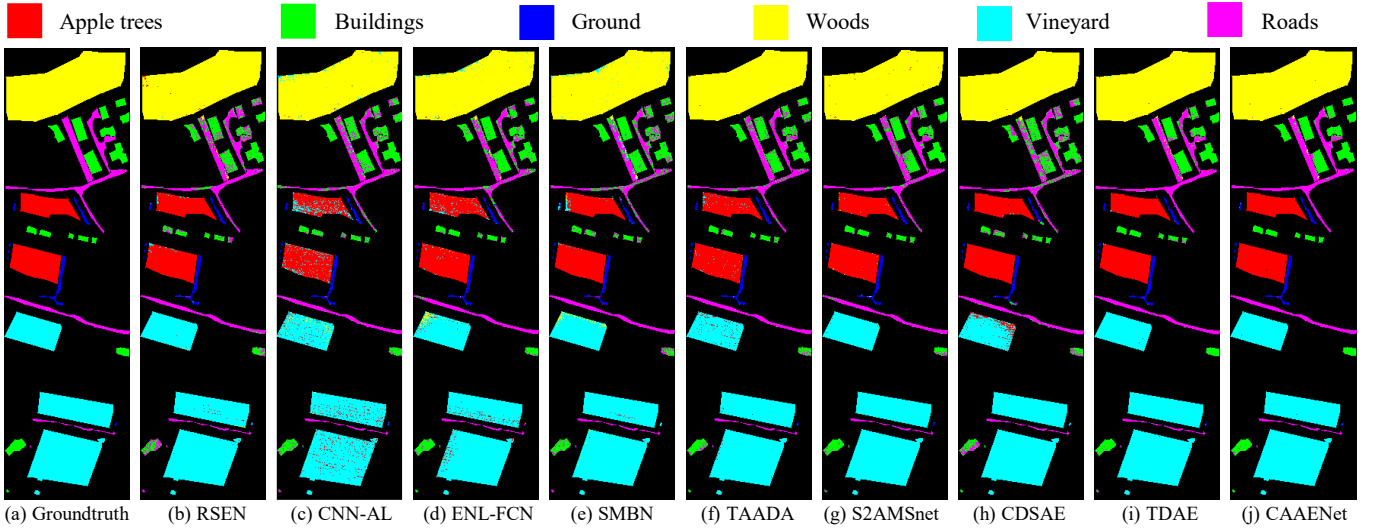


Fig. 4. Classification maps of different methods with 1% ratio of training samples on the Trento data.

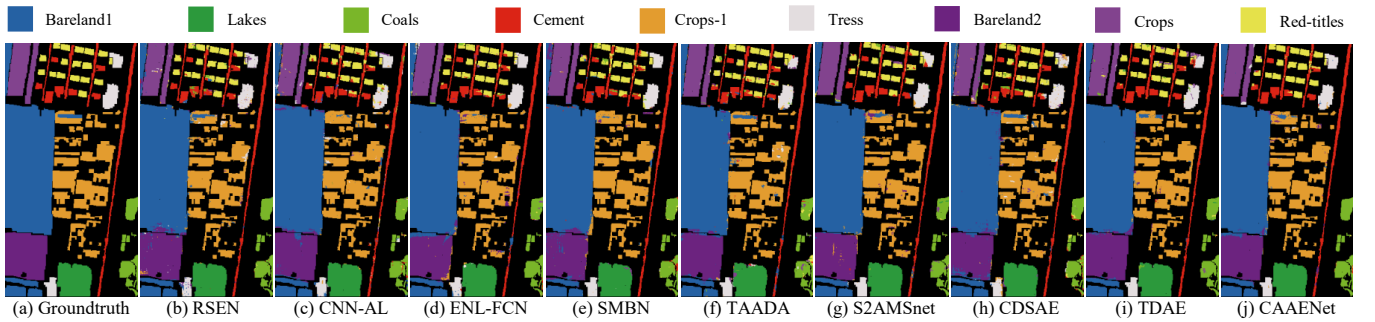


Fig. 5. Classification maps of different methods with 5% ratio of training samples on the Xuzhou data.

FCN [30], SMBN [29], TAADA [50], S2AMSnet [51], CDSAE [36], and TDAE [38]. The RSEN is self-ensembling network based. The CNN-AL, ENL-FCN, and SMBN are based on CNNs. The TAADA and S2AMSnet are constructed with the Adversarial network framework. The CDSAE and TDAE are based on the auto-encoder network.

2) *Implementation Settings*: In the following experiments, some implementation settings are given as follows: For the

proposed CAAENet method for HSI classification, according to Section III-B, the values of both λ_1 and λ_2 are set to 0.1. The values of the learning rates η_P , η_E , and η_D are selected as 10^{-5} , as well as the value of η_W is 10^{-4} , which are determined by the grid search scheme from 10^{-6} to 10^1 . In addition, the parameter values of RSEN, CNN-AL, ENL-FCN, SMBN, TAADA, S2AMSnet, CDSAE, and TDAE are suggested by [39], [28], [30], [29], [50], [51], [36], and [38].

TABLE I
METRIC RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET.

Categories	Classification Methods								
	RSEN	CNN-AL	ENL-FCN	SMBN	TAADA	S2AMSnet	CDSAE	TDAE	CAAENet
Broccoli green weeds_1	97.29	100	99.78	100	99.54	99.77	100	100	100
Broccoli green weeds_2	99.66	99.66	99.65	100	99.88	100	99.65	99.88	99.60
Fallow	99.79	97.86	99.78	99.78	99.30	99.76	100	100	99.75
Fallow rough plow	100	99.69	100	100	100	100	100	99.68	100
Fallow smooth	99.68	99.52	98.69	99.00	100	99.64	98.84	99.50	99.43
Stubble	99.47	99.36	100	99.78	99.77	99.88	99.78	99.78	99.75
Celery	99.77	100	99.64	99.63	99.62	99.87	99.63	99.63	99.58
Grapes untrained	90.82	95.57	94.12	95.78	93.45	94.97	95.35	96.32	97.11
Soil vineyard develop	99.93	99.80	99.44	98.80	100	99.39	99.72	100	100
Corn senesced green weeds	99.23	98.97	98.53	100	98.73	99.86	99.87	99.87	99.39
Lettuce romaine 4 week	96.85	99.61	100	99.59	99.14	100	99.59	99.18	99.53
Lettuce romaine 5 week	100	100	100	100	100	100	100	100	100
Lettuce romaine 6 week	98.62	98.57	98.56	98.62	99.50	99.48	98.08	99.04	99.45
Lettuce romaine 7 week	99.22	97.25	97.17	95.49	99.15	99.56	95.49	99.18	97.21
Vineyard untrained	91.07	91.36	92.11	94.52	91.72	93.87	94.15	94.09	96.54
Vineyard vertical trellis	98.57	99.26	98.26	98.57	100	100	99.00	99.00	99.15
OA(%)	96.49	97.53	97.33	97.97	97.32	97.99	97.92	98.27	98.69
AA(%)	98.12	98.49	98.55	98.70	98.74	99.13	98.70	99.07	99.16
Kappa(%)	96.09	97.24	97.03	97.74	97.01	97.76	97.69	98.08	98.54

TABLE II
METRIC RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET.

Categories	Classification Methods								
	RSEN	CNN-AL	ENL-FCN	SMBN	TAADA	S2AMSnet	CDSAE	TDAE	CAAENet
Apple trees	98.19	83.90	94.84	97.87	96.89	99.34	99.72	99.72	100
Buildings	74.53	84.77	90.27	83.26	86.14	90.16	74.96	94.03	96.43
Ground	95.04	96.26	95.04	88.19	98.38	97.45	84.86	97.45	96.73
Woods	99.62	98.44	98.64	98.65	99.88	99.73	100	99.85	99.93
Vineyard	99.47	93.18	96.50	98.65	98.41	99.88	97.30	99.96	99.97
Roads	86.94	82.35	79.95	75.14	87.50	86.13	77.56	91.39	93.51
OA(%)	95.56	91.63	94.57	94.43	96.33	97.35	94.02	98.39	98.89
AA(%)	92.30	89.82	92.54	90.29	94.53	95.45	89.07	97.07	97.76
Kappa(%)	94.07	88.83	92.75	92.56	95.10	96.46	92.04	97.85	98.52

TABLE III
METRIC RESULTS OF DIFFERENT METHODS ON THE XUZHOU DATASET.

Categories	Classification Methods								
	RSEN	CNN-AL	ENL-FCN	SMBN	TAADA	S2AMSnet	CDSAE	TDAE	CAAENet
Bareland1	97.29	96.33	97.43	95.19	98.53	98.52	97.69	97.93	98.01
Lakes	99.54	98.97	98.60	97.86	95.69	99.66	98.56	98.21	99.92
Coals	98.42	92.03	95.45	93.27	87.59	92.81	87.33	96.00	98.05
Cement	93.19	90.24	91.82	89.46	89.04	92.02	92.89	97.07	93.58
Crops-1	95.71	96.23	96.61	97.70	96.85	97.43	94.83	98.30	97.72
Tress	85.97	83.66	92.70	95.07	83.88	90.13	84.66	86.85	94.91
Bareland2	96.46	95.80	92.20	94.68	95.53	96.64	91.75	97.91	95.68
Crops	96.94	95.64	98.13	98.85	98.68	97.42	95.17	97.91	97.43
Red-titles	88.47	85.01	89.60	86.69	89.37	88.04	89.71	86.57	97.52
OA(%)	95.95	94.78	95.63	95.14	95.54	96.63	94.82	96.97	97.32
AA(%)	94.66	92.66	94.50	94.31	92.76	94.74	92.51	95.20	96.98
Kappa(%)	94.86	93.37	94.45	93.84	94.33	95.72	93.41	96.16	96.60

The details of comparison results and discussion are provided as follows.

3) *Comparison Results*: Fig. 3, Fig. 4, and Fig. 5 are adopted to provide the classification maps of RSEN, CNN-AL, ENL-FCN, SMBN, TAADA, S2AMSnet, CDSAE, TDAE, and CAAENet on the Salinas, Trento, and Xuzhou datasets. As can be found, the classification maps of the proposed CAAENet method achieve the closer groundtruths than those of RSEN, CNN-AL, ENL-FCN, SMBN, TAADA, S2AMSnet, CDSAE, and TDAE. For instance, in Fig. 3, the “Vineyard untrained” regions produced by eight competing approaches have serious errors, while the region yielded by the CAAENet lies in the less errors relatively. In Fig. 4 and Fig. 5, the “Roads” and “Tress” regions for the eight competing approaches have more gap with the regions for ground truths than those for the CAAENet method. Therefore, the proposed CAAENet method obtains more competitive performance of HSI classification compared with other approaches in the qualitative evaluation.

In addition, Table I, Table II, and Table III are adopted to

present the OA, AA, and Kappa results of RSEN, CNN-AL, ENL-FCN, SMBN, TAADA, S2AMSnet, CDSAE, TDAE, and CAAENet on the these three datasets, where each listed value has been averaged over 10 times and the best values have been marked in **bold**. As found from these tables, the OA, AA, and Kappa values of the CAAENet method are all larger than those of the eight competing approaches. In other words, from the perspective of quantitative evaluation, the CAAENet method achieves the better performance than others.

According to the above qualitative and quantitative evaluation, it is concluded that the proposed CAAENet outperforms the RSEN, CNN-AL, ENL-FCN, SMBN, TAADA, S2AMSnet, CDSAE, TDAE, and CAAENet. The main reason is that, different from the compared approaches, our proposed CAAENet method takes full consideration of both feature distribution and separability, so that the highly separable and discriminative features are captured to achieve the high-performance HSI classification.

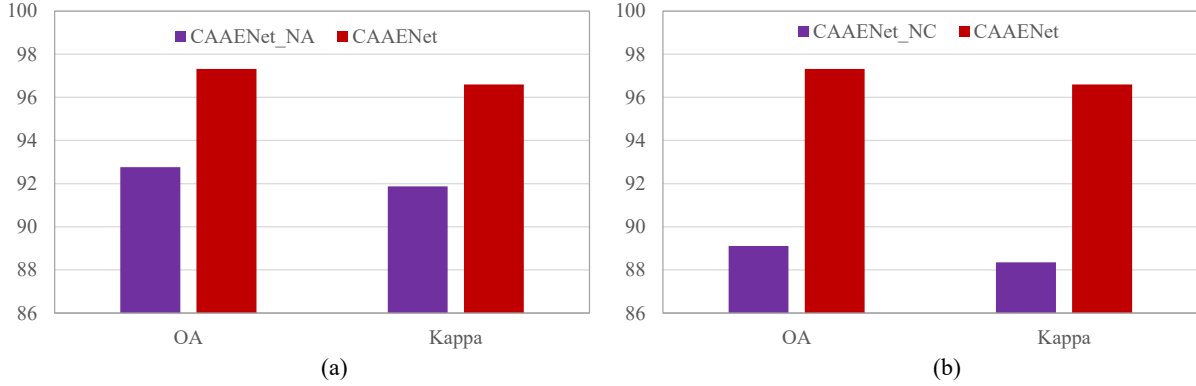


Fig. 6. CAAENet is compared with: (a) CAAENet form without the adversarial loss (denoted by CAAENet_NA); (b) CAAENet form without the cosine loss (denoted by CAAENet_NC). The OA and Kappa metrics are used for result evaluation. The Xuzhou dataset is adopted as testing data.

D. Ablation Study

For more comprehensive exploration of the performance of the proposed CAAENet method, we conduct the ablation study to demonstrate the effectiveness of the adversarial loss and cosine loss. The OA and Kappa metrics are here used for result assessment, and the Xuzhou data is adopted as testing data. The results are visually presented in Fig. 6. The details of the ablation study are shown as follows.

1) *Effectiveness of Adversarial Loss*: The adversarial loss is included in the overall loss (1) to guarantee the feature distributions, which can improve the performance of the proposed CAAENet method. To demonstrate the effectiveness of the adversarial loss, we remove it from the CAAENet and obtain the form without the adversarial loss (denoted by CAAENet_NA). Thus, the CAAENet_NA is compared with the original CAAENet form, with the results shown in Fig. 6(a). As shown in this figure, the CAAENet method obtains the better OA and Kappa results compared with the CAAENet_NA form. The reason is that, once the adversarial loss is removed, we will have no access to acquire the satisfactory feature distribution, so that the classification performance is limited. Therefore, the adversarial loss is highly effective in the CAAENet for the HSI classification.

2) *Effectiveness of Cosine Loss*: The cosine loss is designed in the overall loss (1) to achieve the separability and discriminability for improvement of classification performance. In order to verify the effectiveness of the cosine loss, we remove this loss from (1) so as to obtain the form without the cosine loss (denoted by CAAENet_NC). The CAAENet_NC form is also compared with the CAAENet, with the results shown in Fig. 6(b). From this figure, we can directly see that the OA and Kappa values of CAAENet are bigger than those of CAAENet_NC. In other words, once the cosine loss is removed, it is very difficult to capture the highly separable and discriminative features, which fails to enhance the classification performance. Thence, the cosine loss contributes to the proposed CAAENet method for the HSI classification.

IV. CONCLUSION

In this work, a novel cosine-driven adversarial auto-encoder network (CAAENet) is proposed for HSI classification. To

be more specific, for this CAAENet, the AE network is firstly given to be treated as the basic framework. Then, to increase the distribution consistency between the input HSIs and their features, we construct the adversarial loss in the AE network. With such manner, the distribution difference is continuously reduced by the discriminator during the adversarial training. In addition, the cosine loss is also built into the basic network to improve the separability of features. Thus, the intra-class features are aggregated and the inter-class distance is enlarged, facilitating the classification calculation. Furthermore, we provide an alternating optimization strategy to obtain the optimal solutions of the overall loss of CAAENet. At last, the label prediction of the input HSIs is achieved by imposing the one-hot encoding on a series of the calculation of the obtained solutions. Sufficient experimental results have been provided to demonstrate that the proposed CAAENet method achieves the better performance compared with several advanced approaches in the HSI classification task.

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